

Data Analysis

From a research problem to a research
question to a research design

Luis Carlos Guevara
Faculty of Finance,
Government and International Relations

2026-I

Topic List, Problem Statements, and Purpose Statements

It is crucial to clearly establish a Purpose Statement to provide structure to the research abstract and introduction. This statement **establishes the intent or objective of the research regarding a specific problem. It represents a critical declaration and must be clear, specific, and informative.**

The Purpose Statement conveys the overall intent of the proposed study in one or several sentences. In research proposals, you must clearly distinguish between, the **Research Problem**, the **Research Questions**, and the **Purpose Statement**.

Example Case: consider a topic driven by a specific need, people refusing the COVID-19 vaccine

- **Topic & Problem:** we start with a broad issue. In our example, the topic is the refusal of vaccines, and the problem is the public health risk associated with low vaccination rates.
- **The Research Question:** Narrows the study further by posing one or more specific questions to be answered.
Example: "Do participants refuse vaccination because they are concerned about long-term health consequences?"
Although many factors may influence the decision, this study chooses to specifically address concerns about long-term health benefits.
- **Statement of Purpose:** once the question is set, the Purpose defines how the study will address it. It serves as the bridge between the question and the method.

Academic year 2026 - I

Names:

EXAMPLE		
Topic denomination and Research Problem	Research Questions	Purpose Statement
Public Health & Institutional Trust (COVID-19 Vaccination). Problem: Despite the availability of vaccines, a significant portion of the population refuses to be vaccinated. While the side effects are often cited, there is a gap in understanding how lack of political trust in the government specifically drives this refusal, leading to continued public health risks.	Quantitative: Is there a significant relationship between low levels of trust in the central government and the decision to refuse the COVID-19 vaccine? Qualitative: <i>How do unvaccinated individuals describe the role of government distrust in their health decision-making process?</i>	The purpose of this explanatory sequential mixed-methods study is to first measure the impact of government trust on vaccination rates using a survey, and then to explain these results by exploring the specific narratives of distrust among those who refused the vaccine.
Answer these questions based on the Research Question and Purpose you defined above. Your answers will dictate your final design.		
Causal Inference	To answer your RQ, is it necessary to prove that X <i>caused</i> Y?	YES. We aim to infer if a lack of trust (X) is a predictor/driver of the refusal to vaccinate (Y), rather than just describing that both exist.
Generalization	To answer your RQ, is it necessary to generalize results from your sample to the entire population?	YES. We aim to establish a statistical link between two specific constructs: Government Trust and Vaccination Status. To claim that "Low Trust predicts Refusal" for the public, we must measure both variables at the individual level within the same representative sample. We cannot rely on separate datasets (e.g., one for trust, one for vaccines); we need a single sample to assess the correlation and generalize it to the population by applying a survey.
Change over time	To answer the research question, is there need to examine change over time?	NO. The study aims to identify the relationship between current levels of trust and the decision to vaccinate (which has already happened or is currently being refused). We are taking a Cross-Sectional "snapshot" to see how these variables correlate at this specific moment, rather than tracking how trust fluctuates or evolves over months or years.

Interaction	To answer your RQ, is it necessary to interact directly with participants (talk, observe) rather than just observe data points?	YES. While the survey captures the trend, addressing the explanatory nature of our design requires direct interaction. We cannot fully understand the reasons for distrust through a survey alone. Therefore, in the second phase of the design, we must interact directly (via interviews) with a sub-group of participants to gather the narrative data needed to explain the statistical results found in the first phase.
Data Generation	To answer your RQ, is it necessary to generate your own data (primary) rather than using existing databases (secondary)?	YES. A survey alone gives numbers but lacks depth; interviews alone lack generalizability. We need both to explain the "Impact of Trust."
Multi Method	To answer your RQ, is a single method insufficient? Do you need more than one design?	YES. A single method is insufficient because a quantitative survey can statistically prove that low trust correlates with vaccine refusal, but it cannot fully explain why that specific distrust exists or how it operates in a person's mind. We need a second design (qualitative interviews) to explain the statistical trends found in the first phase.
Design Type	Methods	
Explanatory Sequential Design	• Data sources, sampling ○ Phase 1 (Quant): Online Survey (Google Forms) distributed to a general population (N=100+). Random sampling preferred. ○ Phase 2 (Qual): Purposive sampling. Select 5-10 participants from Phase 1 who answered, "Low Trust" and "Not Vaccinated" for follow-up. • Measurements ○ Quant: Likert Scale (1-5) for "Trust in Gov" + Binary "Vaccinated (Yes/No)". ○ Qual: Semi-structured interview guide ("Can you tell me about when you lost trust...?"). • Analysis ○ Quant: Logistic Regression. ○ Qual: Thematic Analysis of interview transcripts. • Ethics ○ Informed consent stating that survey answers may lead to an invitation for an interview. Anonymity in data storage.	

Problem 1

Problem 1		
Topic denomination and Research Problem	Research Questions	Purpose Statement
Answer these questions based on the Research Question and Purpose you defined above. Your answers will dictate your final design.		
Causal Inference	To answer your RQ, is it necessary to prove that X <i>caused</i> Y?	
Generalization	To answer your RQ, is it necessary to generalize results from your sample to the entire population?	
Change over time	To answer the research question, is there need to examine change over time?	
Interaction	To answer your RQ, is it necessary to interact directly with participants (talk, observe) rather than just observe data points?	

Data Generation	To answer your RQ, is it necessary to generate your own data (primary) rather than using existing databases (secondary)?	
Multi Method	To answer your RQ, is a single method insufficient? Do you need more than one design?	
Design Type	Methods	
Explanatory Sequential Design	• Data sources, sampling • Measurements • Analysis • Ethics	

Problem 2

Problem 1		
Topic denomination and Research Problem	Research Questions	Purpose Statement
Answer these questions based on the Research Question and Purpose you defined above. Your answers will dictate your final design.		
Causal Inference	To answer your RQ, is it necessary to prove that X <i>caused</i> Y?	
Generalization	To answer your RQ, is it necessary to generalize results from your sample to the entire population?	
Change over time	To answer the research question, is there need to examine change over time?	
Interaction	To answer your RQ, is it necessary to interact directly with participants (talk, observe) rather than just observe data points?	

Data Generation	To answer your RQ, is it necessary to generate your own data (primary) rather than using existing databases (secondary)?	
Multi Method	To answer your RQ, is a single method insufficient? Do you need more than one design?	
Design Type	Methods	
Explanatory Sequential Design	• Data sources, sampling • Measurements • Analysis • Ethics	

Problem 3

Problem 1		
Topic denomination and Research Problem	Research Questions	Purpose Statement
Answer these questions based on the Research Question and Purpose you defined above. Your answers will dictate your final design.		
Causal Inference	To answer your RQ, is it necessary to prove that X <i>caused</i> Y?	
Generalization	To answer your RQ, is it necessary to generalize results from your sample to the entire population?	
Change over time	To answer the research question, is there need to examine change over time?	
Interaction	To answer your RQ, is it necessary to interact directly with participants (talk, observe) rather than just observe data points?	

Data Generation	To answer your RQ, is it necessary to generate your own data (primary) rather than using existing databases (secondary)?	
Multi Method	To answer your RQ, is a single method insufficient? Do you need more than one design?	
Design Type	Methods	
Explanatory Sequential Design	• Data sources, sampling • Measurements • Analysis • Ethics	